

Polymer Plant Scheduling — Dedicated PE & PP Production Trains

Manufacturing scheduling case study · Julia + JuMP + HiGHS · companion to `notebooks/polymer_pe_pp_scheduling.jl`

This report documents a **month-long production schedule** (a 30-day, 720-hour horizon) for a polymer plant that manufactures **8 grades of polyethylene (PE)** and **polypropylene (PP)**. Each grade runs as a multi-day campaign, and changeovers between grades take a few hours — the scale real plant schedulers actually work at. PE and PP are made on **physically separate trains**: different reactor technology and catalyst systems (e.g. gas-phase/slurry PE reactors vs. Ziegler-Natta/metallocene PP reactors) mean a PE line can never run a PP grade or vice versa. So this is really **two independent single-line scheduling problems, running in parallel** — one per train.

Within a train, switching between grades still costs time and money: color/additive changes and partial purges. The scheduler below decides the order in which each train runs its confirmed orders so as to minimize total changeover cost plus tardiness cost, per train. Each train's sub-problem is a mixed-integer linear program (MILP), built with [JuMP.jl](#) and solved with the open-source [HiGHS](#) solver.

1. Plant data: grades, families, and rates

Eight grades run on the plant's two trains over a **30-day (720-hour) scheduling horizon**: four PE grades (two HDPE grades, one LLDPE, one LDPE) on the PE train, and four PP grades (two homopolymer variants, one random copolymer, one impact copolymer) on the PP train.

Grade	Train	Rate (t/h)
HDPE-5502	PE	5.0
HDPE-6200	PE	5.5
LLDPE-2020	PE	4.5
LDPE-1922	PE	4.0
PP-Homo-1100	PP	5.5
PP-Homo-1305	PP	5.0
PP-Copo-3300	PP	4.8
PP-Impact-7015	PP	4.5

2. Customer orders → production lots

A plant does not run one production campaign per purchase order. Sales orders for the **same grade** shipping close together get pooled into one production lot, sized to cover all of them, timed to the **earliest** ship date in the group — producing early never hurts the later orders in the group, it just means finished goods sit in the warehouse a few extra hours. Fewer, larger lots also mean fewer changeovers. This run uses a **24-hour** consolidation window (same-grade orders due within that many hours of the earliest one in the group are pooled together).

Customer	Grade	Qty (t)	Due (h)
A1	HDPE-5502	220	212
A2	HDPE-5502	180	220
A3	HDPE-5502	200	548
A4	HDPE-5502	200	556
B1	HDPE-6200	240	260
B2	HDPE-6200	200	268
B3	HDPE-6200	240	596
B4	HDPE-6200	200	604
C1	LLDPE-2020	190	188
C2	LLDPE-2020	150	196
C3	LLDPE-2020	190	524
C4	LLDPE-2020	150	532
D1	LDPE-1922	150	308
D2	LDPE-1922	130	316
D3	LDPE-1922	150	644
D4	LDPE-1922	130	652
E1	PP-Homo-1100	260	236
E2	PP-Homo-1100	210	244
E3	PP-Homo-1100	260	572
E4	PP-Homo-1100	210	580
F1	PP-Homo-1305	220	284
F2	PP-Homo-1305	180	292
F3	PP-Homo-1305	220	620
F4	PP-Homo-1305	180	628
G1	PP-Copo-3300	200	224
G2	PP-Copo-3300	160	232
G3	PP-Copo-3300	200	560
G4	PP-Copo-3300	160	568

Customer	Grade	Qty (t)	Due (h)
H1	PP-Impact-7015	175	332
H2	PP-Impact-7015	140	340
H3	PP-Impact-7015	175	668
H4	PP-Impact-7015	140	676

32 customer orders consolidate into **16 production lots**:

Lot	Grade	Qty (t)	Due (h)	Weight	Combines
1	HDPE-5502	400	212	1.0	2 customer order(s)
2	HDPE-5502	400	548	1.0	2 customer order(s)
3	HDPE-6200	440	260	1.0	2 customer order(s)
4	HDPE-6200	440	596	1.0	2 customer order(s)
5	LLDPE-2020	340	188	1.0	2 customer order(s)
6	LLDPE-2020	340	524	1.0	2 customer order(s)
7	LDPE-1922	280	308	1.0	2 customer order(s)
8	LDPE-1922	280	644	1.0	2 customer order(s)
9	PP-Homo-1100	470	236	1.5	2 customer order(s)
10	PP-Homo-1100	470	572	1.5	2 customer order(s)
11	PP-Homo-1305	400	284	1.5	2 customer order(s)
12	PP-Homo-1305	400	620	1.5	2 customer order(s)
13	PP-Copo-3300	360	224	1.5	2 customer order(s)
14	PP-Copo-3300	360	560	1.5	2 customer order(s)
15	PP-Impact-7015	315	332	1.5	2 customer order(s)
16	PP-Impact-7015	315	668	1.5	2 customer order(s)

3. Sequence-dependent changeovers (within a train)

Because a train only ever runs grades from one family, the only changeover decision is *within* that family. Real purge/transition times depend on the **specific grade pair**, not just "same grade or not" — and they are often **asymmetric**: moving to a lighter color or lower-additive grade purges faster than the

reverse, because residual material from the *previous* grade is what has to be flushed out. Hours shown are from (row) → to (column):

PE train

from \ to	HDPE-5502	HDPE-6200	LLDPE-2020	LDPE-1922
HDPE-5502	0.5	2.0	3.0	6.0
HDPE-6200	2.5	0.5	3.5	6.5
LLDPE-2020	3.5	3.0	0.5	4.0
LDPE-1922	5.0	5.5	4.5	0.5

PP train

from \ to	PP-Homo-1100	PP-Homo-1305	PP-Copo-3300	PP-Impact-7015
PP-Homo-1100	0.5	1.5	3.0	4.0
PP-Homo-1305	2.0	0.5	3.5	4.5
PP-Copo-3300	2.5	3.0	0.5	3.0
PP-Impact-7015	3.5	4.0	2.5	0.5

There is no PE ↔ PP transition to model — that changeover never happens, because it would require re-equipping the whole train.

Cost rates used in this run: changeover **\$450/h**, tardiness **\$300/h** per unit order weight. (The companion Pluto notebook exposes both as live sliders.)

4. MILP formulation (applied once per train)

Each train's orders are scheduled independently. Node 0 is a dummy "train idle" start state; nodes 1..m are that train's orders. Binary variables x_{ij} indicate that order j runs immediately after node i . Continuous variables C_i and T_i are the completion time and tardiness of order i .

```

minimize    c_co * sum_{i != j} s_ij * x_ij  +  c_tar * sum_i w_i * T_i

subject to  sum_j x_0j = 1
            sum_{i != j} x_ij = 1                for all j
            sum_{j != i} x_ij ≤ 1                for all i
            C_j ≥ C_i + s_ij + p_j - M(1 - x_ij)  for all i != j

```

$$T_i \geq C_i - d_i, \quad T_i \geq 0$$

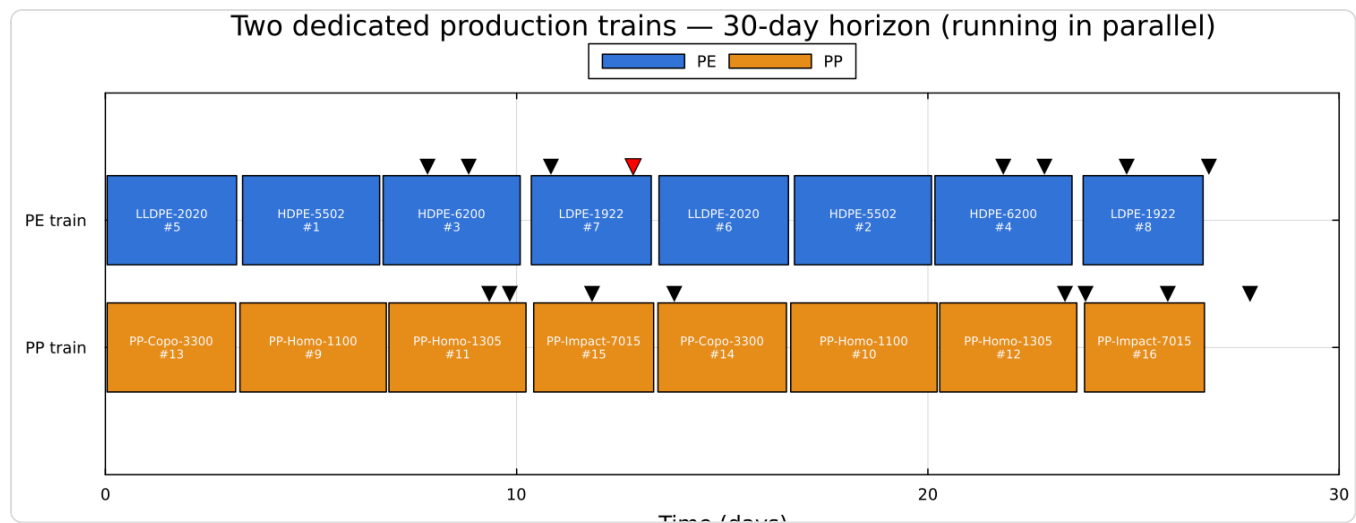
$$x_{ij} \in \{0,1\}, \quad C_i, T_i \geq 0$$

Completion times must strictly increase along any chain of active arcs, so this "big-M timing" formulation rules out sub-tours on its own — no separate subtour-elimination constraints (e.g. MTZ) are needed. The two trains' sub-problems share no variables, so solving them independently with HiGHS is exact, not a heuristic decomposition — both reach global optimality in well under a second.

5. Optimal campaign sequence, per train

Train	Order	Grade	Qty (t)	Start (day)	End (day)	Due (day)	Status
PE	5	LLDPE-2020	340	0.0	3.2	7.8	on time
PE	1	HDPE-5502	400	3.3	6.7	8.8	on time
PE	3	HDPE-6200	440	6.8	10.1	10.8	on time
PE	7	LDPE-1922	280	10.4	13.3	12.8	0.4 d late
PE	6	LLDPE-2020	340	13.5	16.6	21.8	on time
PE	2	HDPE-5502	400	16.8	20.1	22.8	on time
PE	4	HDPE-6200	440	20.2	23.5	24.8	on time
PE	8	LDPE-1922	280	23.8	26.7	26.8	on time
PP	13	PP-Copo-3300	360	0.0	3.2	9.3	on time
PP	9	PP-Homo-1100	470	3.3	6.8	9.8	on time
PP	11	PP-Homo-1305	400	6.9	10.2	11.8	on time
PP	15	PP-Impact-7015	315	10.4	13.3	13.8	on time
PP	14	PP-Copo-3300	360	13.4	16.6	23.3	on time
PP	10	PP-Homo-1100	470	16.7	20.2	23.8	on time
PP	12	PP-Homo-1305	400	20.3	23.6	25.8	on time
PP	16	PP-Impact-7015	315	23.8	26.7	27.8	on time

6. Gantt chart



Grade blocks are colored by polymer family (**PE** vs. **PP**); triangles mark each order's due date (red = missed). Because the two trains never compete for the same equipment, they run **simultaneously**: overall makespan is the *slower* of the two trains, not the sum of both.

7. Key performance indicators

PE TRAIN MAKESPAN	PP TRAIN MAKESPAN	OVERALL MAKESPAN	TOTAL COST
26.7 d	26.7 d	26.7 d	\$25667

Tardiness: 0.4 d across 1 of 16 orders. Slack vs. the 30-day horizon: 3.3 days. PE train cost: \$16442; PP train cost: \$9225.

8. Discussion & extensions

Modeling PE and PP as two **dedicated, parallel trains** — rather than one shared line that pays a family-changeover penalty — matters economically: it removes a changeover that can never actually happen on real equipment, and it lets both product families progress at once instead of queueing behind each other. Natural next steps for a production-grade version:

- **More than one train per family** — e.g. two PE lines with different capability sets; extend the model to a proper parallel-machine assignment problem within a family.

- **Minimum/maximum campaign length** — avoid uneconomically short runs by bounding the number of consecutive orders of the same grade.
- **Data-driven changeover matrix** — populate the grade-pair matrix from real historical purge-time records (e.g. a CSV export from the plant historian) instead of hand-entered estimates.
- **Shared upstream/downstream utilities** — if both trains draw from a common feedstock or packaging line, that shared resource needs its own capacity constraint linking the two schedules.
- **Rolling-horizon re-optimization** — re-solve as new orders arrive or priorities change, warm-starting from the current sequence.
- **Robustness** — sample uncertain processing rates / late raw-material arrivals and re-optimize, or solve a chance-constrained variant.

Because each train's model is a compact, independent JuMP program, all of the above are incremental changes to the scheduling function above, not a rewrite. An interactive version of this same model — with live cost-weight sliders — is available as a Pluto.jl notebook at `notebooks/polymer_pe_pp_scheduling.jl` in this repository.